

U20CAM-IMX662-S1 UserManual



1. General Description:

The Innomaker U20CAM-IMX662-S1 is a compact USB 2.0 UVC camera module featuring the Sony STARVIS 2 IMX662 1/2.8-inch CMOS image sensor with $2.9\ \mu\text{m}$ pixels, designed to deliver excellent image quality in low-light environments. It supports Full HD 1920x 1080 video at up to 60 fps in MJPEG format, as well as YUY2 output, and features Clear HDR/DOL HDR for improved image detail in challenging lighting conditions. Equipped with a 135° diagonal and 111° horizontal wide-angle lens, the camera provides a broad field of view suitable for surveillance, robotics, machine vision, and embedded vision applications. Its native UVC compatibility enables plug-and-play operation without additional drivers on Windows, Linux, Ubuntu, macOS, and Android, while supporting easy integration with PCs, Raspberry Pi, Jetson Nano, and other industrial SBCs. With a compact 32 x 32 mm PCBA, four M2 mounting holes, and built-in ESD and EMI protection for high-speed differential signal lines, the U20CAM-IMX662-S1 provides a reliable and flexible imaging solution for both development and OEM applications.

2. Features

1.[Native UVC Compliance] High-Speed USB 2.0 Interface, Native UVC driver on Windows 11/10/7, Mac OS, Linux, Ubuntu and Android system. Direct integration with Raspberry Pi, Jetson Nano, Notebook, Desktop and industrial SBCs.

2.[Superior Performer] On-board 1/2.8 inch IMX662 sensor, Speed up to 1920x1080@60fps(MJPEG). Support YUY2 and MJPEG format. FOV (D) = 135° , FOV(H) = 111° .

3.[Performance for Low Light] Sony STARVIS 2 sensor with $2.9\ \mu\text{m}$ pixels and Clear-HDR//DOL HDR captures crisp, low-noise images down to $<0.1\ \text{lux}$. Enhanced NIR sensitivity ensures reliable night vision with IR illuminators—ideal for 24/7 surveillance and low-light monitoring.

4.[OEM-Ready Compact Design] 32x32mm PCBA size, with 4 x M2 holes. Matching enclosure ASIN: B0CLXV25NR.

5.[Enhanced Protection] Built-in ESD (electrostatic discharge) and EMI (electromagnetic interference) protection for high-speed differential signal lines, ensuring stable performance and reduced signal noise in various environments.

3. Technical Specification

Sensor	Name	IMX662-AAQR-C
	Size	1/2.8 Inch
	Resolution	1920*1080
	Image Color	Colours
	Shutter type	rolling shutter
Lens	F(N) /Aperture	2.0±10%
	Focal Length	3±5%
	TV DISTORTION	<20%
	CRA	15.3 Degree
	Lens Diameter	M12
	Lens Seat Spacing	20mm
	FOV-D	134.8 Degree
	FOV-H	111 Degree
PCBA	Size	32*32 mm
	Layers	4
	Mounting Holes	4 counts, Diameter 2.2mm
	Trigger	None
	Strobe Flash	None
	Night vision	Yes
	Microphone	Yes
	ESD/EMI	Yes
Interfac es	Communication Interface	USB 2.0 High-Speed
	Cable Length	1 M
Software	Device Name	U20CAM-IMX662-S1
	VID/PID	0x0BDA /0x636B
	Communication Standard	UVC 1.0.0
	Supported format	YUY2, MJPEG

	Supported Resolution Supported Frame rate	MJPES Output resolutions: 1920*1080 @60fps 1280x960 @60fps 1280x720 @60fps 1024x768 @60fps 960x540 @60fps 800x600 @60fps 640x480 @60fps 640x360 @60fps 352x288 @60fps 320x240 @60fps YUY2 Output resolutions: 1920*1080 @ 5fps 1280x960 @ 8fps 1280x720 @10fps 960x540 @20fps 800x600 @20fps 640x480 @30fps
	Auto Parameters	White Balance Exposure
	Controllable Parameters	Brightness, Contrast, Hue, Saturation, Sharpness, Gamma, White Balance, Backlight Comp, Gain, Exposure, PowerLine Frequency, Low Light Compensation
	Support System	Windows , Linux , Mac Os, Android

3. Software

UVC (USB Video Class) is a protocol standard for USB cameras, allowing for plug-and-play functionality on various platforms. UVC cameras are widely used because they support high-definition video and work across many operating systems without the need for additional drivers.

<https://github.com/INNO-MAKER/UVC-Camera-Software>

You can refer to the official UVC documentation below, but please note that not all UVC cameras are based on version 1.5. You should check the version you are using and refer to the corresponding documentation.

<https://www.usb.org/document-library/video-class-v15-document-set>

4. Case

We designed a aluminum alloy case for the camera module. You could find it on our Amazon shop. If you can't catch it, feel free to let me know.

<https://github.com/INNO-MAKER/U20CAM-CASE>



5. User Manual Version Descriptions

Version	Description	Date	E-mail
V1.0		2026.07.01	support@inno-maker.com sales@inno-maker.com

If you have any suggestions, ideas, codes and tools please feel free to email to me. I will update the user manual and record your name and E-mail in list. Look forward to your letter and kindly share.